



Euclidean Geometry In Mathematical Olympiads

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P1.49(USAMO 2010/P1)



USAMO

Chapter 1

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Solution

Let T be the projection of Y onto AB , by simson line P, Q, T and R, S, T are collinear, so $T = PQ \cap RS$, then $\angle YTA = \angle YPA = 90^\circ \implies ATYP$ cyclic, similarly, we have $BTYS$ cyclic, so

$$\angle PTS = \angle PTY + \angle YTS = \angle PAY + \angle YBS = \frac{\widehat{XZ}}{2}, \text{ since } O \text{ is center of semicircle, } \angle XOZ = \widehat{XZ}, \text{ so we are done.}$$

This post has been edited 1 time. Last edited by illogical_21, Mar 17, 2019, 7:30 PM

kootrapali

4527 posts

Mar 3, 2020, 1:00 PM

#13

Solution

Let T be the foot of the altitude from Y to AB . Notice that since Y lies on (AXB) and (AZB) and P, Q, T lie on the Simson line from Y to $\triangle AXB$ and S, R, T lie on the simson line from Y to $\triangle AZB$. Thus, the acute angle formed by PQ and RS is $\angle PTS$. Since $\angle YPA = \angle YTA = 90^\circ$ and $\angle YSB = \angle YTB = 90^\circ$, quadrilaterals $ATYP$ and $BTYS$ are cyclic. Finally,

$$\angle PTS = \angle PTY + \angle YTS = \angle PAY + \angle YBS = \angle XAY + \angle YBZ = \frac{1}{2}\angle XOY + \frac{1}{2}\angle YOZ = \frac{1}{2}\angle XOZ.$$

Jerry122805

1074 posts

Aug 14, 2020, 4:48 PM

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Solution

First note that lemma 1.49 tells us the intersection of SR and PQ lies on AB and call that intersection T . Then $YT \perp AB$. Now note that $YPAT$ and $YSBT$ are both cyclic. Now

$$\angle QTY = \angle PTY + \angle YTZ = \angle PAY + \angle YBS = \frac{1}{2}(\angle XTY + \angle YTZ) = \frac{1}{2}\angle XOY.$$

and we are done.

Solution with directed angles

We proceed the same with lemma 1.49. Now since $\angle YPA = \angle YTA$ we have Y, P, A, T are concyclic. Now $\angle QIY = \angle PIY = \angle PAY = \frac{1}{2}\angle XIY$ and